Development Environment Setup and First Steps

Aims

Set up the development environment for Symfony and create a first web application

Objectives

- Install and understand all required software/tools
- Create a new skeleton application
- Understand the structure of Symfony web applications
- Create a first route
- Create a first controller

Requirements and recommended tools

PHP

Version 8.4 or newer, non-thread-safe is usually okay

Windows:

- https://www.php.net/downloads.php (https://www.php.net/downloads.php)

- extract and copy to a folder such as C:\PHP\php8.4

- add to path variable

- copy php.ini_production to php.ini

- php.ini: extension_dir = "ext"

- php.ini: extension = openssl

(- php.ini: error_log = C:\PHP\logs\errors.log after creating a logs directory in PHP)

php -v to check version that is being used

php -i to check entire configuration

Composer
https://getcomposer.org/ (https://getcomposer.org/)

documentation: https://getcomposer.org/doc/ (https://getcomposer.org/doc/)

Symfony CLI
https://symfony.com/download (https://symfony.com/download)

Executable (may require admin rights)

check requirments (symfony check:req)

create new Symfony project (symfony new project_name)

run a development web server (symfony serve or symfony server:start)

PhpStorm
IDE with Symfony support

On campus - installed in computer labs under Windows and Linux

Linux - may require configuration of activation/licensing server:

http://uos-p-lics-05.isdads.salford.ac.uk:1006

To use on your own computer:

Download and install from https://www.jetbrains.com/phpstorm/ (https://www.jetbrains.com/phpstorm/)

Create a free student account to get access to JetBrains software: https://www.jetbrains.com/community/education/#students (https://www.jetbrains.com/community/education/#students)

- Use JetBrains account when asked for license

install Symfony plugins

configure composer

Symfony projects

Continuous improvement of Symfony – new versions coming out regularly

Potential for compatibility problems between major versions (changes in setup/configuration, folder structure, best practice etc) and difficulties with dependencies (which might not work with all versions of Symfony)

https://symfony.com/releases (https://symfony.com/releases)

Create a new project
Framework sources/dependencies are installed based on a composer.json config file

Easiest way: symfony new project_name

(use option --no-git if you do not have git installed)

Alternatively: composer create-project symfony/website-skeleton project_name

Forcing a specific PHP version: php7.2 /usr/local/bin/composer create-project symfony/website-skeleton project_name

--version to create a project using a specific Symfony version (default latest, x.4 typically LTS version)

Recommended: symfony new my_project_name --version=lts

Folder structure and purpose of elements
/bin - location of bin/console

/config - configuration files

/public - location of front controller (web root)

/src - main location for your code (controllers etc)

/var - cache, logs etc

/vendor - framework sources and any other external packages

composer.json - composer configuration file (list of dependencies including any constraints on what updates to apply)

composer.lock - exact record of the dependency versions that have been installed

symfony.lock - lock file for recipes (Flex)

bin/console

Symfony projects come with an additional tool: bin/console

Swiss army knife for Symfony development

php bin/console to see all options

e.g. bin/console debug:router

can be extended by other packages/own extensions

Run Symfony Applications

Start a webserver

Easiest way: Symfony executable (symfony serve in project root)

Alternatively, any web server – pointing to the public folder of the Symfony project

e.g. from within the project root: php -S 127.0.0.1:8000 -t public/

Open a browser, navigate to 127.0.0.1:8000

Implement a first web page

Create a route and a controller for an about page:

Route

in config/routes.yaml

```
about:

   path: /about

   controller: App\Controller\DefaultController::about
```

Controller

in src/Controller/DefaultController.php

```
<?php

namespace App\Controller;

use Symfony\Component\HttpFoundation\Response;

class DefaultController
```

```
{

   public function about()

   {

       return new Response('About page – produced by my first Symfony
controller');

   }

}
```